

Serial Interfaces

Lecture 11

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Outline

- Serial Interfaces Overview
 - Advantages over parallel
 - Major considerations
 - Overview of protocols
- Serial Peripheral Interface
 - Description
 - MCU configuration
- CMSIS device header
- SPI demo
- DS1722 SPI temperature sensor
 - Datasheet overview

Learning Objectives

By the end of this lecture you should be able to...

- See how the SPI peripheral works on the STM32L432KC
- See how to verify the output using a logic analyzer
- Use the CMSIS device header ([stm32l432xx.h](#)) to access peripheral registers in a driver

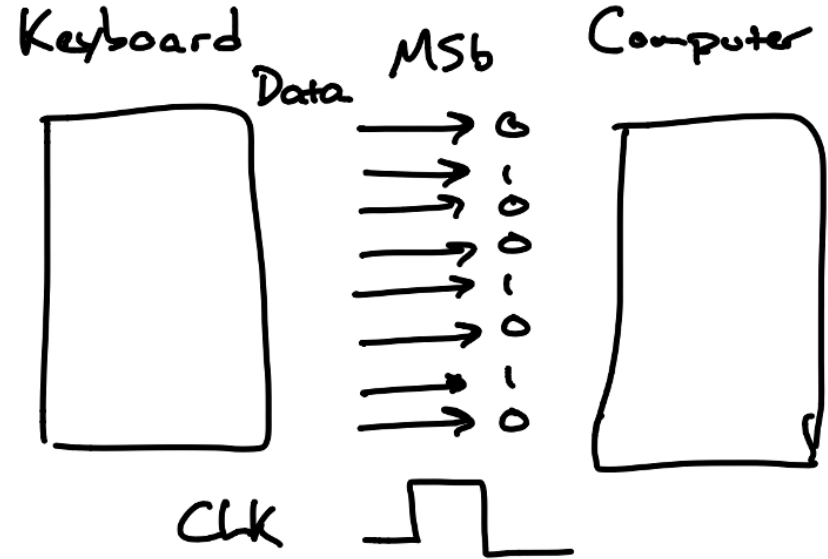
Serial Interfaces Overview

Motivation

How can we interface a peripheral?

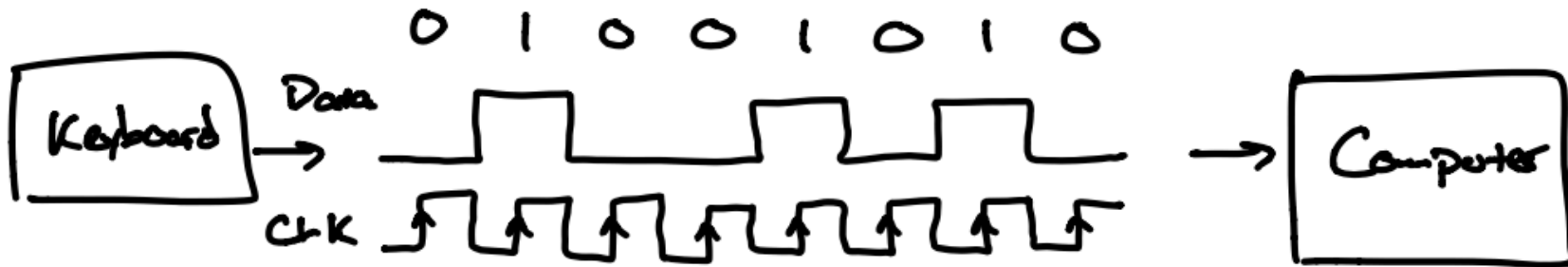
Imagine transmitting a character on a keyboard.

Capital J in ASCII is $74_{10} = 01001010_2$



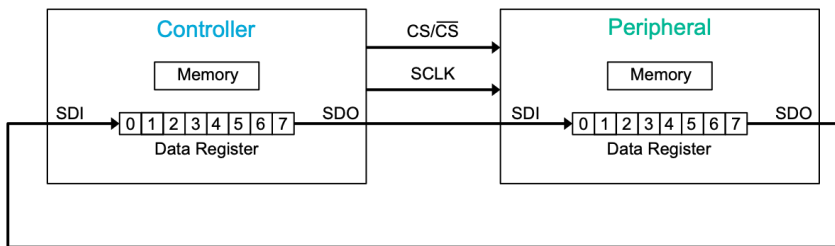
What if we repackage data in a stream?

- Multiplexing in time
- To send N bits, we only need 2 lines (CLK + Data) instead of 9
- Price we pay is time – but often worth it.



Serial Peripheral Interface (SPI)

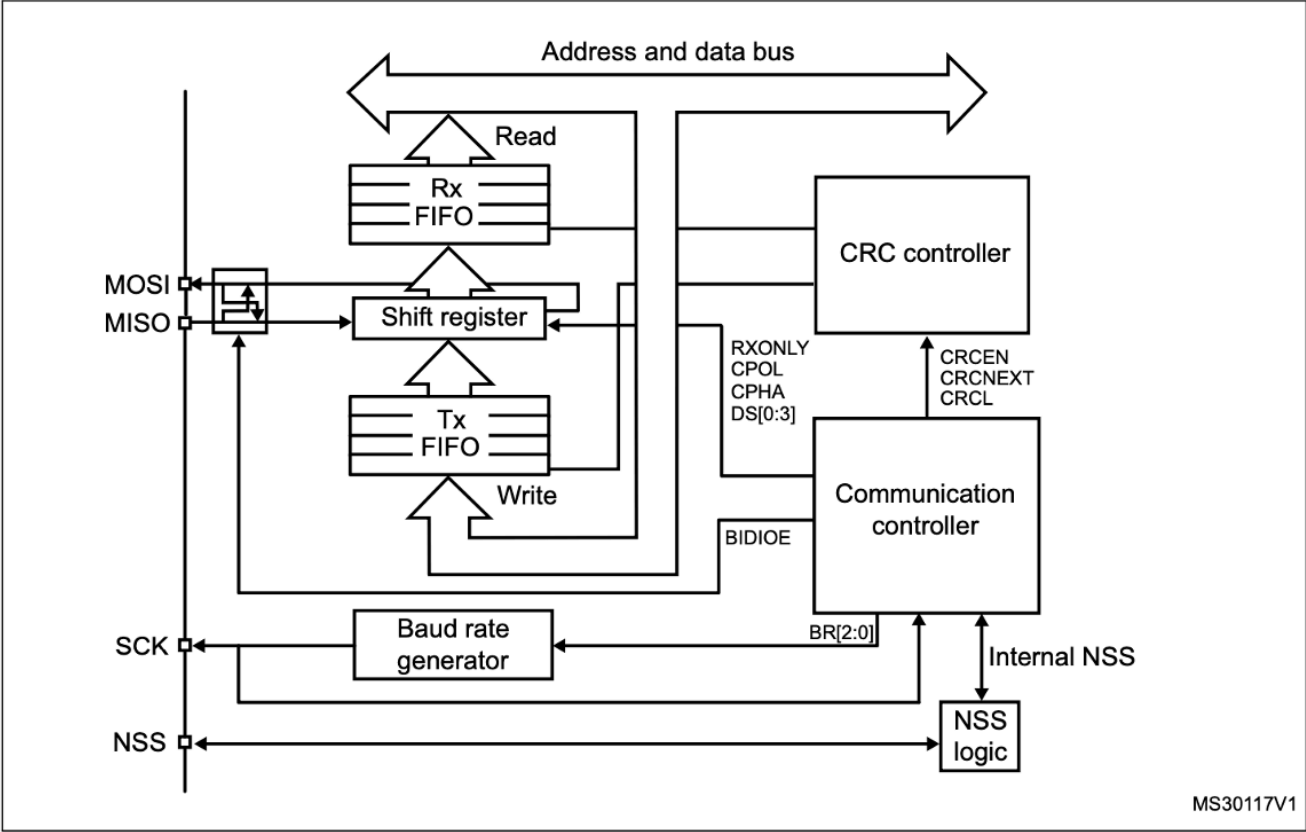
- Developed in the mid-1980s by Motorola
- Used to interface with many peripherals like memory (SD cards, flash), displays, sensors (accelerometers, gyroscopes, temperature sensors, ADCs and DACs).
- Four-wire, synchronous serial bus



SCLK: Serial clock
MOSI: Master Out Slave In
MISO: Master In Slave Out
CE/CS/nCE/nCS: Chip select/enable

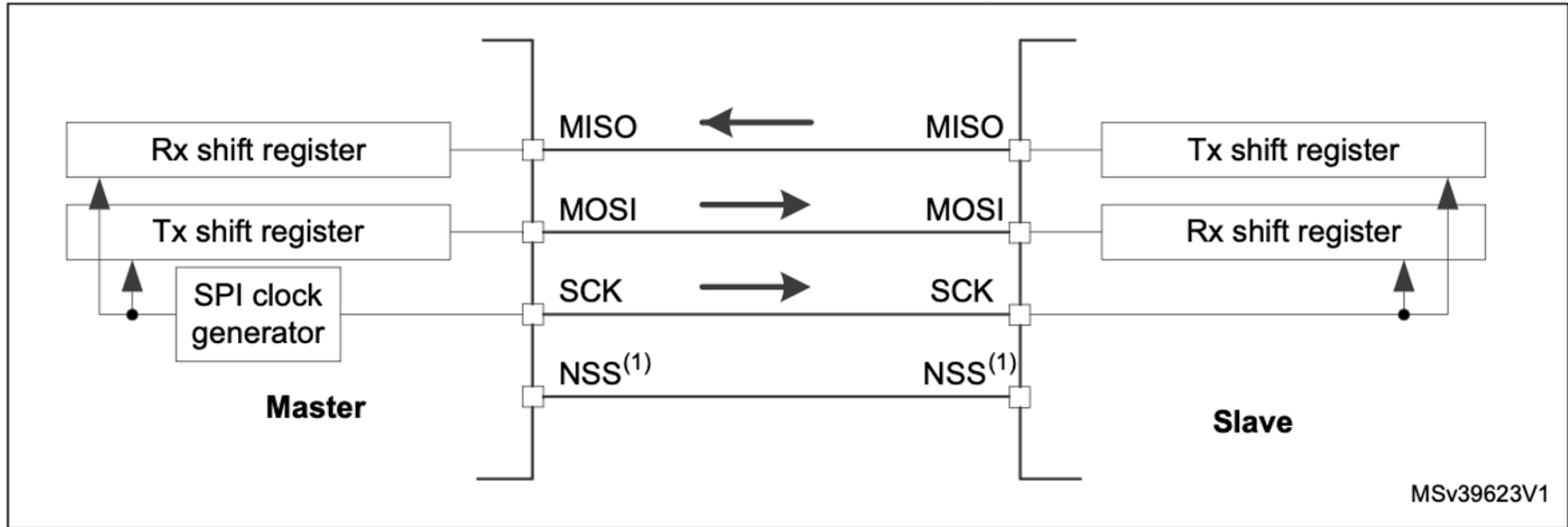
SPI Block Diagram on STM32L432KC

Figure 419. SPI block diagram

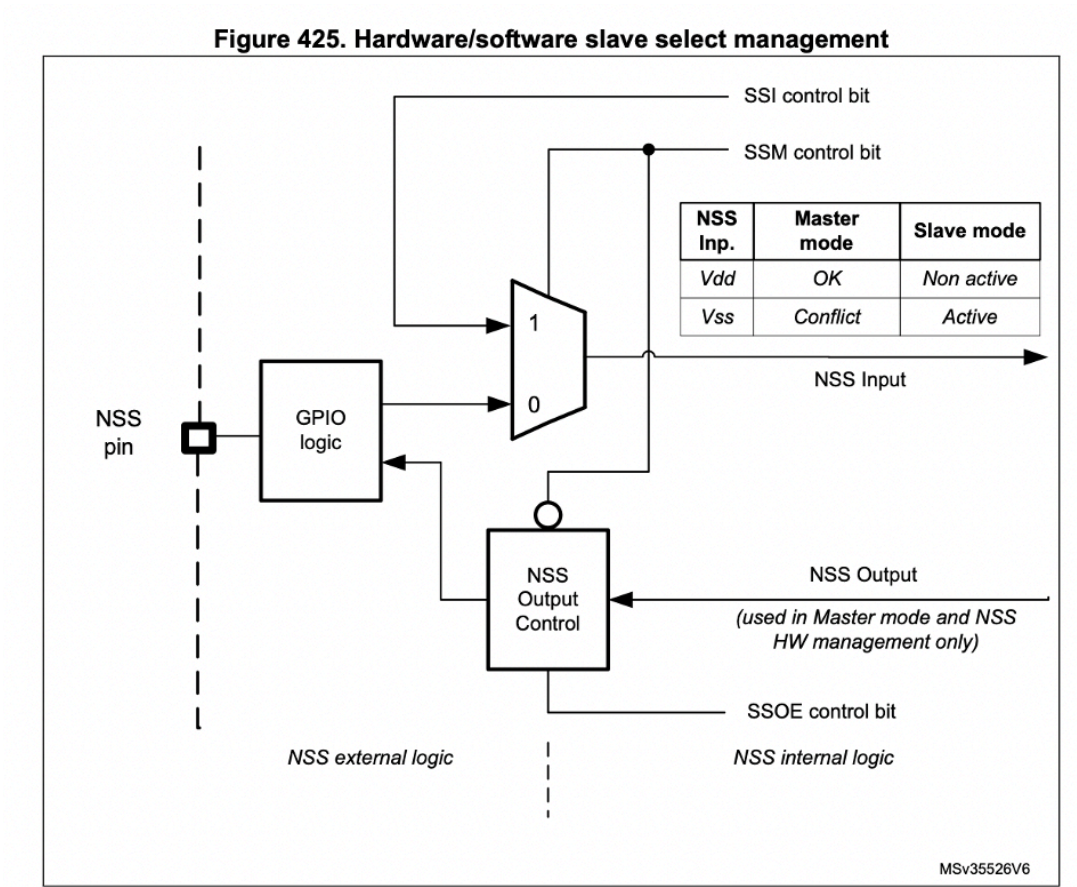


SPI Block Diagram

Figure 420. Full-duplex single master/ single slave application

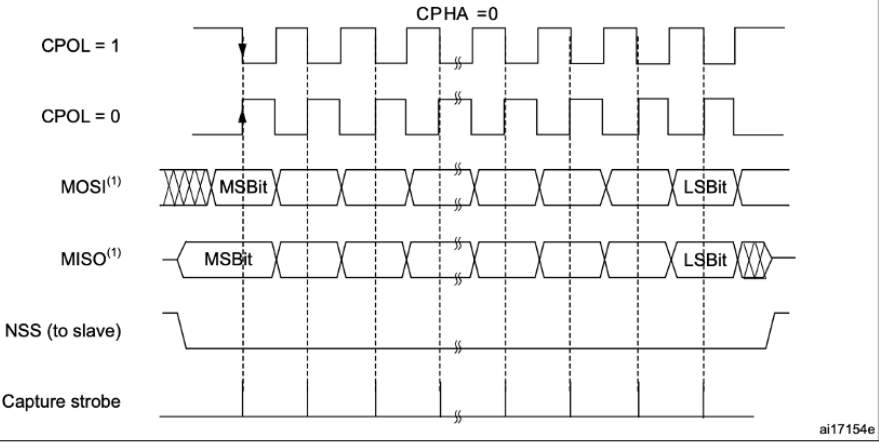
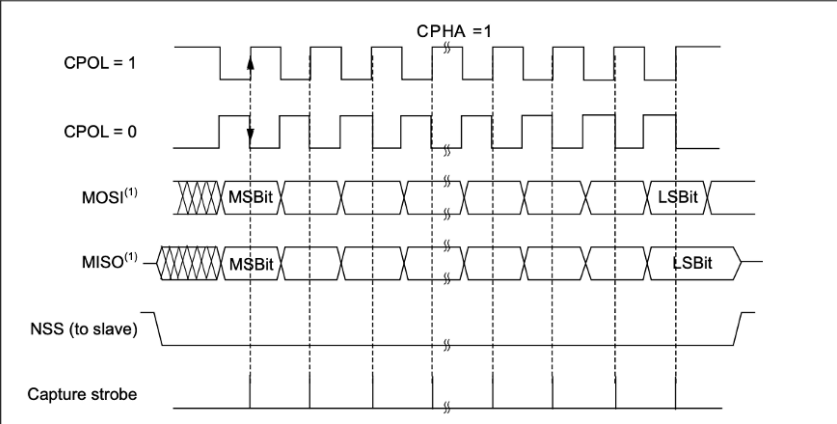


SPI Hardware NSS Management



Example SPI Traces

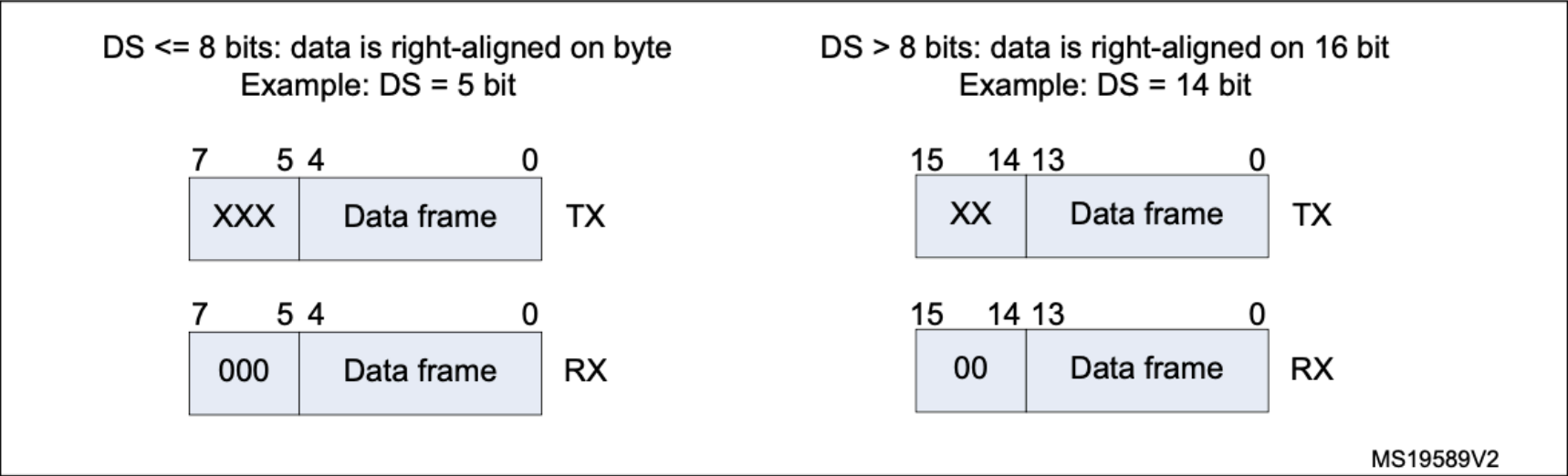
Figure 426. Data clock timing diagram



RM0394. p. 1312

SPI Data Frame Sizes

Figure 427. Data alignment when data length is not equal to 8-bit or 16-bit



The minimum data length is 4 bits. If a data length of less than 4 bits is selected, it is forced to an 8-bit data frame size.

SPI Configuration

Configuration of SPI

The configuration procedure is almost the same for master and slave. For specific mode setups, follow the dedicated sections. When a standard communication is to be initialized, perform these steps:

1. Write proper GPIO registers: Configure GPIO for MOSI, MISO and SCK pins.
2. Write to the SPI_CR1 register:
 - a) Configure the serial clock baud rate using the BR[2:0] bits (Note: 4).
 - b) Configure the CPOL and CPHA bits combination to define one of the four relationships between the data transfer and the serial clock (CPHA must be cleared in NSSP mode). (Note: 2 - except the case when CRC is enabled at TI mode).
 - c) Select simplex or half-duplex mode by configuring RXONLY or BIDIMODE and BIDIOE (RXONLY and BIDIMODE can't be set at the same time).
 - d) Configure the LSBFIRST bit to define the frame format (Note: 2).
 - e) Configure the CRCL and CRCEN bits if CRC is needed (while SCK clock signal is at idle state).
 - f) Configure SSM and SSI (Notes: 2 & 3).
 - g) Configure the MSTR bit (in multimaster NSS configuration, avoid conflict state on NSS if master is configured to prevent MODF error).
3. Write to SPI_CR2 register:
 - a) Configure the DS[3:0] bits to select the data length for the transfer.
 - b) Configure SSOE (Notes: 1 & 2 & 3).
 - c) Set the FRF bit if the TI protocol is required (keep NSSP bit cleared in TI mode).
 - d) Set the NSSP bit if the NSS pulse mode between two data units is required (keep CHPA and TI bits cleared in NSSP mode).
 - e) Configure the FRXTH bit. The RXFIFO threshold must be aligned to the read access size for the SPIx_DR register.
 - f) Initialize LDMA_TX and LDMA_RX bits if DMA is used in packed mode.
4. Write to SPI_CRCPR register: Configure the CRC polynomial if needed.
5. Write proper DMA registers: Configure DMA streams dedicated for SPI Tx and Rx in DMA registers if the DMA streams are used.

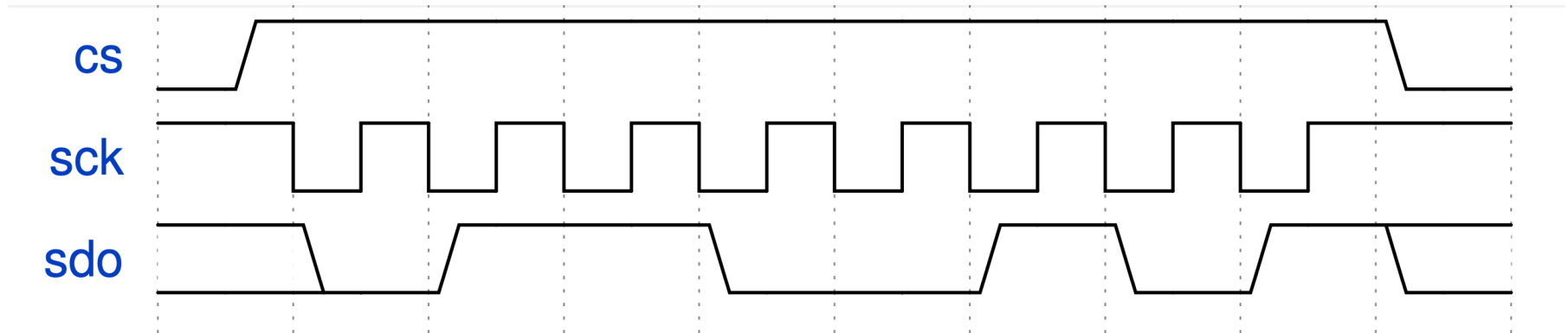
SPI Clock Polarity and Phase

- Clock polarity (CPOL) refers to the state of the clock line at idle
 - 0: clock is **low** when idle
 - 1: clock is **high** when idle
- Clock phase (CPHA) refers to when data is sampled vs. when new data is shifted out
 - 0: the **first** clock transition is the first data capture edge
 - 1: the **second** clock transition is the first data capture edge
- The clock transition (rising or falling) depends on the clock **polarity**
- 4 combinations or modes (CPOL,CPHA) = (0,0), (0,1), (1,0), (1,1)
- Must pay attention to match this mode to the peripheral!

SPI Example 1

Determine the following properties for the waveform below:

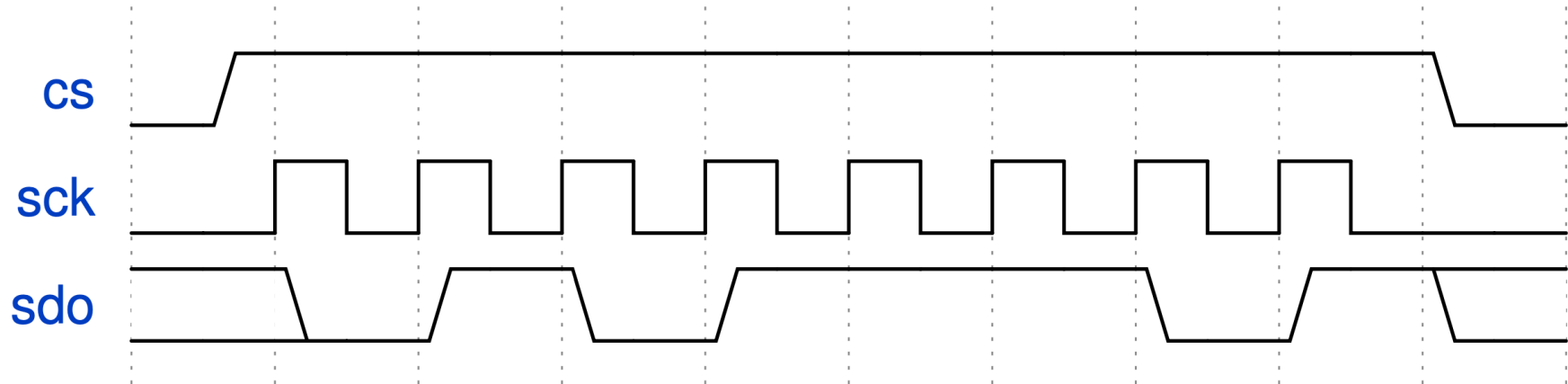
1. Clock Polarity: **1**
2. Clock Phase: **1**
3. Data Packet Value: **0x65**



SPI Example 2

Determine the following properties for the waveform below:

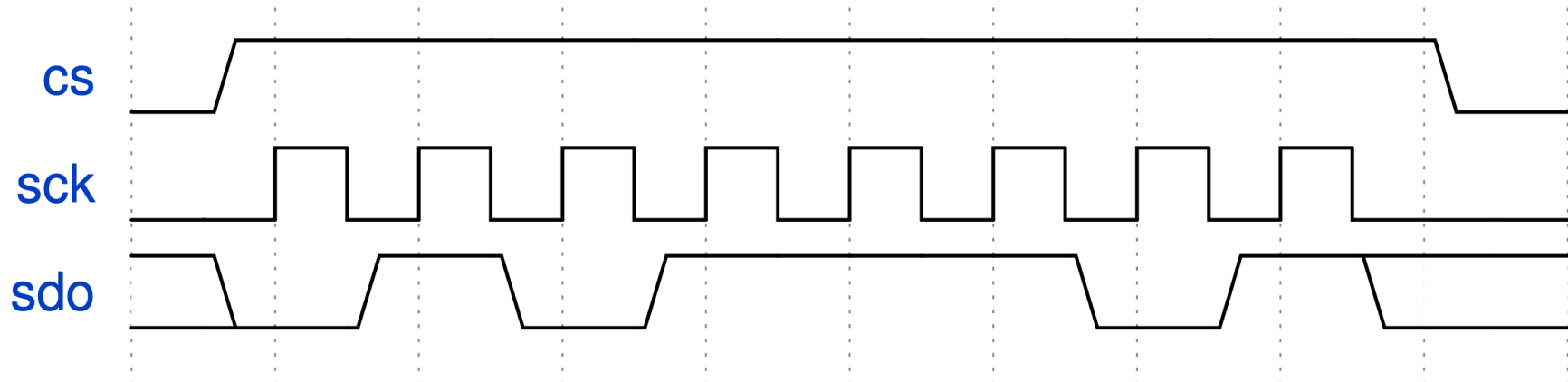
1. Clock Polarity: **0**
2. Clock Phase: **1**
3. Data Packet Value: **0x5d**



SPI Example 3

Determine the following properties for the waveform below:

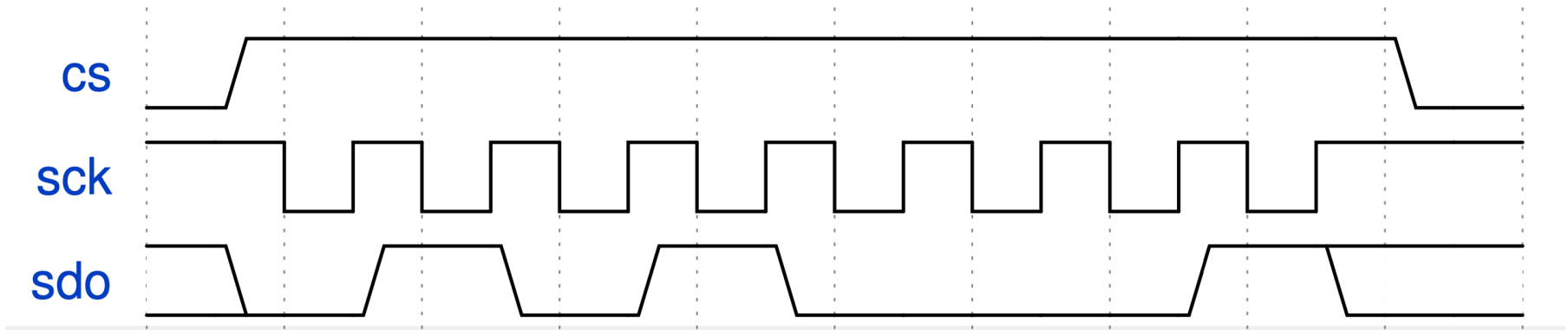
1. Clock Polarity: 0
2. Clock Phase: 0
3. Data Packet Value: 0x5d



SPI Example 4

Determine the following properties for the waveform below:

1. Clock Polarity: **1**
2. Clock Phase: **0**
3. Data Packet Value: **0x51**



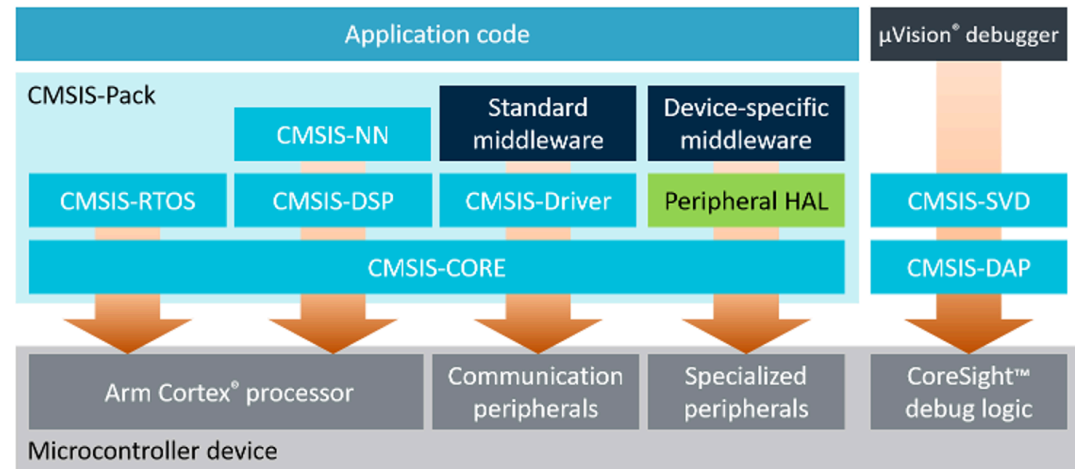
Basic Configuration in Master/Controller Mode

- Configure clock tree
- Turn on SPI clock domain
- Set SPI parameters
 - Clock rate using baud rate divisor
 - CPOL and CPHA to match slave
 - DS[3:0] in CR2 to set the data frame size (4 to 16 bits, usually 8)
 - Set LSBFIRST bit to set whether lsb or msb is sent first (normally msb)
 - Configure the NSS pin (can either use software management or a separate GPIO set as an output and manually toggle it)
 - Set to master mode MSTR
- Enable SPI – Set SPE bit to 1

Common Microcontroller Software Interface Standard (CMSIS)

What is CMSIS?

- Arm's standard software interface for Cortex-M chips
- We only use CMSIS-Core: the device header `stm32l432xx.h` (from ST)
- Same names as the reference manual, so the RM is your guide to the header
- Lab 6: your SPI driver must use it



What's in `stm32l432xx.h`?

1. Base addresses of each peripheral
2. A struct type for each peripheral's registers
3. Bit definitions for every field in every register

```
1 #define PERIPH_BASE      (0x40000000UL)
2 #define APB2PERIPH_BASE  (PERIPH_BASE + 0x00010000UL)
3 #define SPI1_BASE        (APB2PERIPH_BASE + 0x3000UL)
4
5 #define SPI1              ((SPI_TypeDef *) SPI1_BASE)
```

Register Structs

```
1 typedef struct
2 {
3     __IO uint32_t CR1;      // Address offset: 0x00
4     __IO uint32_t CR2;      // Address offset: 0x04
5     __IO uint32_t SR;       // Address offset: 0x08
6     __IO uint32_t DR;       // Address offset: 0x0C
7     __IO uint32_t CRCPR;    // Address offset: 0x10
8     __IO uint32_t RXCRCR;   // Address offset: 0x14
9     __IO uint32_t TXCRCR;   // Address offset: 0x18
10 } SPI_TypeDef;
```

- `__IO` is `volatile`
- `SPI1->CR2` is the word at address `0x40013004`

Bit Definitions

Every field gets a position (`_Pos`) and a mask (`_Msk`)

```
1 #define SPI_CR1_MSTR_Pos  (2U)
2 #define SPI_CR1_MSTR_Msk  (0x1UL << SPI_CR1_MSTR_Pos) // 0x00000004
3 #define SPI_CR1_MSTR      SPI_CR1_MSTR_Msk
4
5 #define SPI_CR1_BR_Pos    (3U)
6 #define SPI_CR1_BR_Msk   (0x7UL << SPI_CR1_BR_Pos)   // 0x00000038
7 #define SPI_CR1_BR       SPI_CR1_BR_Msk
```

Use the mask for single bits, and the position to shift in a field value

```
1 SPI1->CR1 |= SPI_CR1_MSTR;           // Set one bit
2 SPI1->CR1 &= ~SPI_CR1_BR;             // Clear a field...
3 SPI1->CR1 |= (br << SPI_CR1_BR_Pos); // ...then write its value
```

SPI Demo

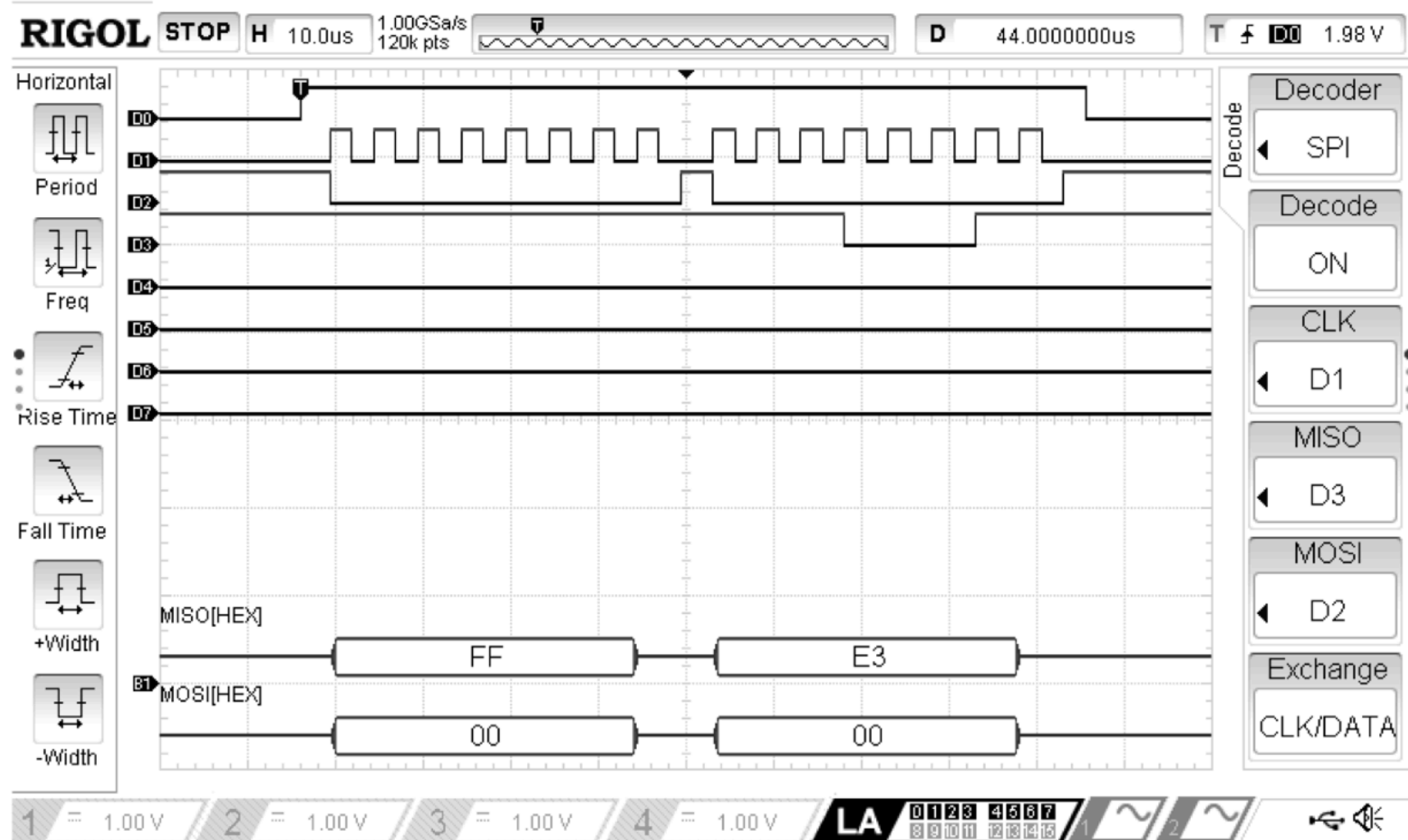
spi_demo.c: main()

```
1  int main(void) {
2      configureFlash();
3      configureClock();
4
5      gpioEnable(GPIO_PORT_A);
6      gpioEnable(GPIO_PORT_B);
7      gpioEnable(GPIO_PORT_C);
8
9      RCC->APB2ENR |= (RCC_APB2ENR_TIM15EN);
10     initTIM(TIM15);
11
12     initSPI(2, 0, 0);
13
14     while(1) {
15         digitalWrite(PA8, PIO_HIGH);
16         spiSendReceive(0xAB);
17         digitalWrite(PA8, PIO_LOW);
18         delay_millis(TIM15, 10);
19     }
20 }
```

SPI.h Function Prototypes

```
1  /* Enables the SPI peripheral and initializes its clock speed (baud rate), polarity, and phase.
2  *    -- br: (0b000 - 0b111). The SPI clk will be the master clock / 2^(BR+1).
3  *    -- cpol: clock polarity (0: inactive state is logical 0, 1: inactive state is logical 1).
4  *    -- cpha: clock phase (0: the first clock transition is the first data capture edge,
5  *                                     1: the second clock transition is the first data capture edge)
6  * Refer to the datasheet for more low-level details. */
7  void initSPI(int br, int cpol, int cpha);
8
9  /* Transmits a character (1 byte) over SPI and returns the received character.
10 *    -- send: the character to send over SPI
11 *    -- return: the character received over SPI */
12 char spiSendReceive(char send);
```

An Example SPI Transaction



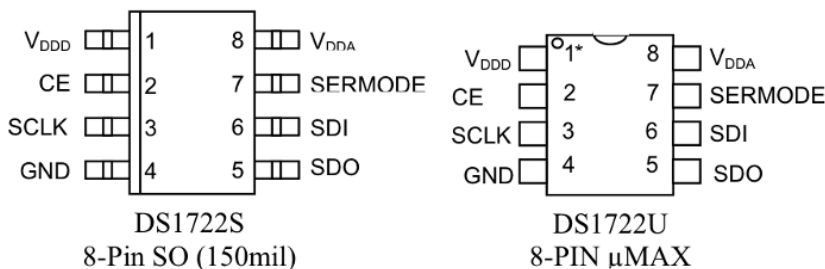
DS1722 SPI Temperature Sensor



DS1722 Digital Thermometer with SPI/3-Wire Interface

www.maxim-ic.com

PIN ASSIGNMENT

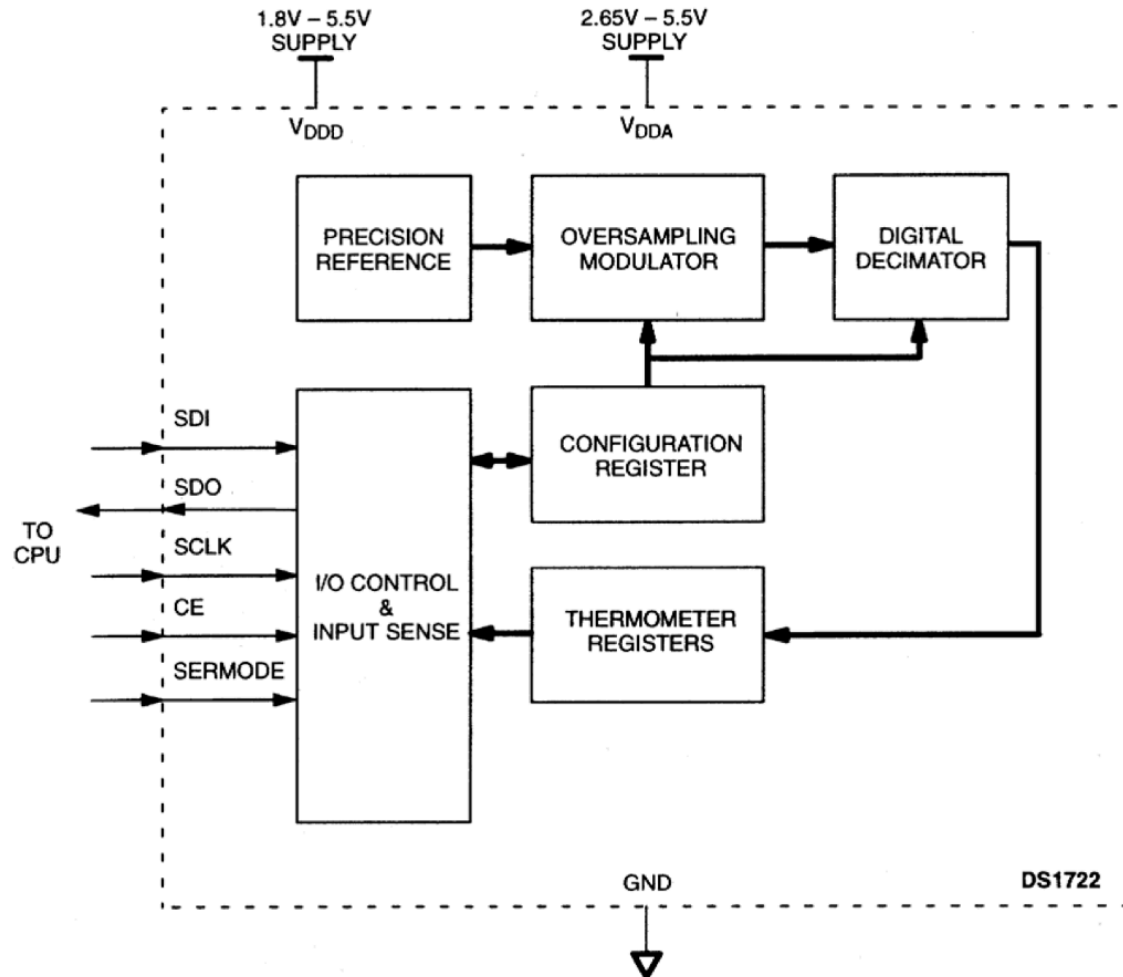


DETAILED PIN DESCRIPTION Table 2

SOIC	SYMBOL	DESCRIPTION
PIN 1	V_{DD}	Digital Supply Voltage 1.8V-5.5V. Defines the top rails for the digital inputs and outputs.
PIN 2	CE	Chip Enable Must be asserted high for communication to take place for either the SPI or 3-wire interface.
PIN 3	SCLK	Serial Clock Input Used to synchronize data movement on the serial interface for either the SPI or 3-wire interface.
PIN 4	GND	Ground pin.
PIN 5	SDO	Serial Data Output When SPI communication is selected, the SDO pin is the serial data output for the SPI bus. When 3-wire communication is selected, this pin must be tied to the SDI pin (the SDI and SDO pins function as a single I/O pin when tied together.)
PIN 6	SDI	Serial Data Input When SPI communication is selected, the SDI pin is the serial data input for the SPI bus. When 3-wire communication is selected, this pin must be tied to the SDO pin (the SDI and SDO pins function as a single I/O pin when tied together.)
PIN 7	SERMODE	Serial Interface Mode Input This pin selects which interface standard will be used: SPI when connected to V_{CC} ; standard 3-wire when connected to GND.
PIN 8	V_{DDA}	Analog Supply Voltage 2.65V – 5.5V input power pin.

DS1722 Functional Block Diagram

DS1722 FUNCTIONAL BLOCK DIAGRAM Figure 1



Temperature Data Register Format

Temperature/Data Relationships Table 3

								Address Location
S	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	02h
MSb		(unit = °C)				LSb		
2 ⁻¹	2 ⁻²	2 ⁻³	2 ⁻⁴	0	0	0	0	01h

TEMPERATURE	DIGITAL OUTPUT (BINARY)	DIGITAL OUTPUT (HEX)
+120°C	0111 1000 0000 0000	7800h
+25.0625°C	0001 1001 0001 0000	1910h
+10.125°C	0000 1010 0010 0000	0A20h
+0.5°C	0000 0000 1000 0000	0080h
0°C	0000 0000 0000 0000	0000h
-0.5°C	1111 1111 1000 0000	FF80h
-10.125°C	1111 0101 1110 0000	F5E0h
-25.0625°C	1110 0110 1111 0000	E6F0h
-55°C	1100 1001 0000 0000	C900h

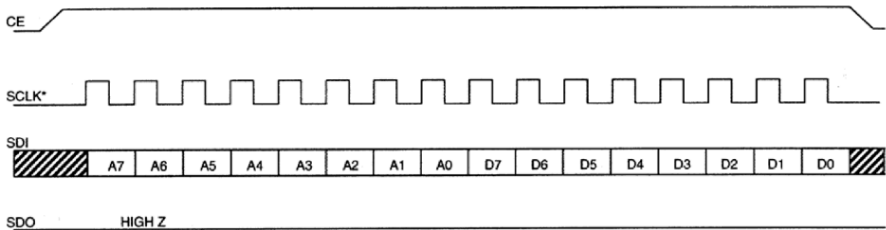
SPI Transactions

Register Address Structure Table 4

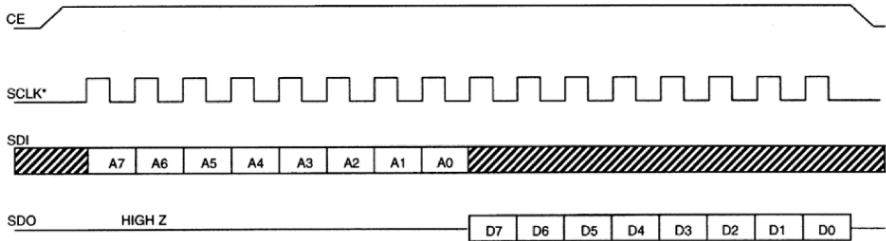
Read Address	Write Address	Active Register
00h	80h	Configuration
01h	No access	Temperature LSB
02h	No access	Temperature MSB

Register structure

SPI SINGLE BYTE WRITE Figure 4



SPI SINGLE-BYTE READ Figure 5



Example waveforms